THE OPEN MAPPING LEMMA FOR COMPLETE LINEARLY TOPOLOGIZED GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Linearly topologized abelian group with a countable fundamental system of neighbourhoods of zero, and completeness. the discussion of the same chapter, source lines 8996-9022; the neighbouring Baire category statement is counted by the sibling card S1-algebra-baire-category-complete-module at 9037-9075. Supplies the nested subgroups N_n, the completeness used to sum the y_i, and the openness of the closures of the images.
- Nowhere dense subset and the failure of the second alternative. used at proof lines 9105-9106 to conclude that the closure of u(N_n) is open. Is the negated hypothesis from which the whole construction starts.
- Separatedness of the target and completeness of the source in the metric-like sense of Cauchy sequences in a filtered group. used at proof lines 9118-9119, where the infinite sum of the y_i is formed and then identified with x_1. Makes the element y exist and makes the equation x_1 = u(y) valid, which is the crux of the proof.

Source reference. Original lemma 0CQW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15976–15976. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2 (Open mapping lemma)**.** *Let $u : N \to M$ be a continuous map of topologized abelian groups. Assume that M separated and that N is complete, linearly topologized, and has a countable fundamental system of neighbourhoods of 0. Then exactly one of the following holds*

(1) *u is open, or*
(2) *for some open subgroup $N' \subset N$ the image $u(N')$ is nowhere dense in M.*

Proof. Let N_n, $n \in \mathbf{N}$ be a fundamental system of neighbourhoods of 0. We may assume that N_n is a subgroup and that $N_{n+1} \subset N_n$. If (2) does not hold, then the closure M_n of $u(N_n)$ is an open subgroup for $n = 1, 2, 3, \ldots$. Since u is continuous, we see that M_n, $n \in \mathbf{N}$ must be a fundamental system of open neighbourhoods of 0 in M. Also, since M_n is the closure of $u(N_n)$ we see that

$$u(N_n) + M_{n+1} = M_n$$

for all $n \geq 1$. Pick $x_1 \in M_1$. Then we can inductively choose $y_i \in N_i$ and $x_{i+1} \in M_{i+1}$ such that

$$u(y_i) + x_{i+1} = x_i$$

The element $y = y_1 + y_2 + y_3 + \ldots$ of N exists because N is complete. Whereupon we see that $x_1 = u(y)$ because M is separated. Thus $M_1 = u(N_1)$. In exactly the same way the reader shows that $M_i = u(N_i)$ for all $i \geq 2$ and we see that u is open. $\square$